Alex Morehead

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ACADEMIC RESEARCH

International Computer Science Institute (ICSI) — **Berkeley, CA**
Affiliated Postdoctoral Researcher (Advisor: Dr. Michael Mahoney) — Apr 2026 – Present

Lawrence Berkeley National Laboratory (LBNL) — **Berkeley, CA**
Hopper Postdoctoral Fellow, Computing Sciences (Advisor: Dr. Michael Mahoney) — Jun 2025 – Present

University of Cambridge — **Remote**
Visiting Research Collaborator, Labs of Profs. Pietro Liò & Sir Tom Blundell — Mar 2023 – May 2025

University of Missouri — **Columbia, MO**
Graduate Research Assistant, Bioinformatics & Machine Learning Lab — Aug 2020 – May 2025

Indiana University-Purdue University Indianapolis — **Indianapolis, IN**
Undergraduate Research Assistant, NSF Data Science REU — Jun 2019 – Aug 2019

Missouri Western State University — **Saint Joseph, MO**
Undergraduate Research Assistant, NSF Synthetic Biology REU — May 2018 – Aug 2018

INDUSTRY RESEARCH

Profluent Bio — **Berkeley, CA**
Research Intern, AI Research Lab (Supervisor: Dr. Jeffrey Ruffolo) — May 2023 – Aug 2023

AACE Venture Fund — **Columbia, MO**
Investment Analyst, Technology Deal Flow Group — Aug 2022 – May 2023

Absci — **New York City, NY**
Research Intern, AI Research Lab (Supervisor: Joshua Meier) — Jun 2022 – Apr 2023

EDUCATION

University of Missouri — **Columbia, MO**
PhD & MS, Computer Science — Aug 2020 – May 2025

- Dissertation: Geometric deep learning and generative modeling of 3D biomolecules
- Collaborations: University of Cambridge, Research Labs of Prof. Pietro Liò & Prof. Sir Tom Blundell
- Advisor: Dr. Jianlin Cheng
- O'Neill and College of Engineering Dean's Graduate Fellow; Highest Distinction (GPA: 4.0/4.0)

Missouri Western State University — **Saint Joseph, MO**
BS, Computer Science — Aug 2016 – May 2020

- Minor in Mathematics
- Graduated top of class among all 2020 graduates in computer science, mathematics, and physics (GPA: 4.0/4.0)
- General Studies and Outstanding Graduate Honors

TEACHING

Lawrence Berkeley National Laboratory — **Berkeley, CA**
Guest Lecturer, Deep Learning for Science (DL4Sci) School — Summer 2026

University of Missouri — **Columbia, MO**
Guest Lecturer, Computational Modeling of Molecular Structures (CMP_SC 8170) — Fall 2024
Guest Lecturer, Advanced Methods in Deep Learning (CMP_SC 8890) — Fall 2022

EnCircle Technologies — **Columbia, MO**
Teaching Assistant, Programming & Reinforcement Learning Curriculum — Summer 2022, 2023

Missouri Western State University — **Saint Joseph, MO**
Content Tutor, Object Oriented Programming & Data Structures (CSC 254 & 285) — Spring 2017 – 2018

JOURNAL

[12] **Morehead***, Atanackovic*, Hegde*, Wang*, Boadu*, Selvaraj*, Tong, Krishnapriyan, Cheng, "Flow matching for generative modeling in bioinformatics and computational biology," *Nature Machine Intelligence*, 2026.

[11] **Morehead***, Liu*, Neupane*, Cheng, "Artificial intelligence methods for protein structure and interaction prediction: Recent advances and challenges," *Current Opinion in Structural Biology*, invited issue on AI Methodologies in Structural Biology, 2026.

[10] Anand*, Joshi*, **Morehead**, et al., "RNA-FrameFlow for de novo 3D RNA Backbone Design," *Transactions on Machine Learning Research (TMLR)*, presented at ICML 2024 SPIGM (*oral*) and AI4Science (*spotlight*) workshops, 2025.

[9] **Morehead**, Giri, Liu, Neupane, Cheng, "Assessing the potential of deep learning for protein-ligand docking," *Nature Machine Intelligence*, also presented as an ICML 2024 AI4Science *spotlight* (top 20% – 30/159), 2025.

[8] **Morehead**, Liu, Neupane, Giri, Cheng, "Protein-ligand structure and affinity prediction in CASP16 using a geometric deep learning ensemble and flow matching," *Proteins: Structure, Function, and Bioinformatics*, presented as a CASP16 top-5 method, 2025.

[7] **Morehead**, Cheng, "Geometry-complete diffusion for 3D molecule generation and optimization," *Communications Chemistry*, 7(1):150, also presented at ICLR 2023 MLDD (poster), 2024.

[6] **Morehead**, Cheng, "Geometry-complete perceptron networks for 3D molecular graphs," *Bioinformatics*, 40(2):btae087, also presented at AAAI 2023 DLG (poster) and AI2ASE (*oral*), 2024.

[5] Chen, Chen, **Morehead**, Wu, Cheng, "3D-equivariant graph neural networks for protein model quality assessment," *Bioinformatics*, 39(1):btad030, 2023.

[4] Lensink et al. (including **Morehead**), "Impact of AlphaFold on structure prediction of protein complexes: The CASP15-CAPRI experiment," *Proteins: Structure, Function, and Bioinformatics*, 2023.

[3] **Morehead**, Chen, Sedova, Cheng, "DIPS-Plus: The enhanced database of interacting protein structures for interface prediction," *Scientific Data*, 10(1):509, 2023.

[2] **Morehead**, Cheng, "Protein Structure Accuracy Estimation using Geometry-Complete Perceptron Networks," *Protein Science*, 33(3):e4932, 2023.

[1] Kouckya, Wagnerb, Aguilerab, Bashawb, Chena, Eckdahla, Edmanc, Gomeza, Hanlanb, Kempfd, Mattoon, McKlin, Mazariegos, **Morehead**, et al., "Synthetic Biology Bicistronic Designs Support Gene Expression Equally Well in vitro and in vivo," *American Journal of Undergraduate Research (AJUR)*, 2020.

CONFERENCE AND WORKSHOPS

[16] La, Gupta, **Morehead**, Cheng, Zhang, "MegaFold: System-Level Optimizations for Accelerating Protein Structure Prediction Models," *ISC High Performance*, 2026.

[15] **Morehead***, Cretu*, Panescu*, Anand*, Weiler*, Perez*, Blau, Farrell, Bhimji, Jain, Sahasrabuddhe, Liò, Jaakkola, Gómez-Bombarelli, Ying, Erichson, Mahoney, "Zatom-1: Towards a Multimodal Foundation Model for 3D Molecules and Materials," *ICLR Workshop on Foundation Models for Science*, under review with an ML venue, 2026.

[14] Ta, **Morehead**, Erichson, Fioretto, Madireddy, "Choosing the Right Representations: Task-Aware Fusion for Multimodal Materials Language Models," *NeurIPS AI for Accelerated Materials Design (AI4Mat) Workshop*, 2026.

[13] Elhag*, Raja*, **Morehead***, Blau, Morris, Bronstein, "Learning Inter-Atomic Potentials without Explicit Equivariance," *NeurIPS Machine Learning in Structural Biology (MLSB) Workshop*, under review with an ML venue, 2025.

[12] Joshi, Jamasb, Viñas, Harris, Mathis, **Morehead**, Anand, Liò, "gRNAde: Geometric Deep Learning for 3D RNA Inverse Design," *International Conference on Learning Representations (ICLR)*, selected as an ICLR 2025 *spotlight*, with follow-up work achieving wet-lab validation, 2025.

[11] **Morehead**, Cheng, "FlowDock: Geometric Flow Matching for Generative Protein-Ligand Docking and Affinity Prediction," *Intelligent Systems for Molecular Biology (ISMB)*, presented as a CASP16 top-5 method, 2025.

[10] Pourmirzaei, **Morehead**, Esmaili, Ren, Pourmirzaei, Xu, "GCP-VQVAE: A Geometry-Complete Language for Protein 3D Structure," *NeurIPS AI4Science Workshop*, under review with an ML venue, 2025.

[9] Jamasb*, **Morehead***, Joshi*, Zhang*, et al., "Evaluating Representation Learning on the Protein Structure Universe," *International Conference on Learning Representations (ICLR)*, also at NeurIPS 2023 MLSB (poster), 2024.

[8] Chen*, **Morehead***, Liu, Cheng, "A gated graph transformer for protein complex structure quality assessment and its performance in CASP15," *Intelligent Systems for Molecular Biology (ISMB)*, 2023.

[7] **Morehead**, Bhatnagar, Ruffolo, Madani, "Towards Joint Sequence-Structure Generation of Nucleic Acid and Protein Complexes," *NeurIPS Machine Learning in Structural Biology (MLSB) Workshop*, 2023.

[6] **Morehead***, Chantapakul*, Cheng, "Semi-Supervised Graph Learning Meets Dimensionality Reduction," *IEEE International Conference on Machine Learning and Applications (ICMLA)*, 2023.

[5] Soltanikazemi*, Roy*, Quadir, Giri, **Morehead**, Cheng, "DRLComplex: Reconstruction of protein quaternary structures using deep reinforcement learning," *International Conference on Intelligent Biology and Medicine (ICIBM)*, 2023.

[4] **Morehead**, Chen, Cheng, "Geometric Transformers for Protein Interface Contact Prediction," *International Conference on Learning Representations (ICLR)*, 2022.

[3] Shoman*, Aboah*, **Morehead**, Duan, Daud, Adu-Gyamfi, "A Region-Based Deep Learning Approach to Automated Retail Checkout," *Proceedings of the IEEE/CVF CVPR Workshops*, 2022.

[2] Gao, Lund-Andersen, **Morehead**, Mahmud, Chen, Chen, Giri, Roy, Quadir, Effler, Prout, Abraham, Elwasif, Haas, Skolnick, Cheng, Sedova, "High-Performance Deep Learning Toolbox for Genome-Scale Prediction of Protein Structure and Function," *IEEE/ACM Machine Learning with Graphs in High Performance Computing Environments (MLHPC) Workshop*, 2021.

[1] **Morehead**, Ogden, Magee, Hosler, White, Mohler, "Low Cost Gunshot Detection using Deep Learning on the Raspberry Pi," *IEEE International Conference on Big Data*, 2019.

PREPRINTS

[5] Cretu*, Abrudan, Panescu, Perez, Anand, Erichson, Mahoney, Blau, Jacobson, Gómez-Bombarelli, Ying, Knowles, Liò, **Morehead***, "Zatom-2: Multitask Pretraining on Atomistic Data for Generative Modeling across Domains," *arXiv*, under review with an ML venue; serving as senior author and project co-lead, 2026. arXiv:2610.11454.

[4] Wang, Jamasb, Hajij, **Morehead**, Braithwaite, Liò, "Topotein: Topological Deep Learning for Protein Representation Learning," *arXiv*, recipient of the University of Cambridge's Computer Science and Technology Highly Commended MPhil Project Prize (2024-2025) awarded to Wang, 2025. arXiv:2509.03885.

[3] Mahmud, **Morehead**, Cheng, "Accurate prediction of protein tertiary structural changes induced by single-site mutations with equivariant graph neural networks," *bioRxiv*, 2023.

[2] Shanehsazzadeh et al. (including **Morehead**), "Unlocking de novo antibody design with generative artificial intelligence," *bioRxiv*, demonstrating wet-lab validation and presenting follow-up work at NeurIPS 2023 MLSB (poster), 2023.

[1] **Morehead**, Chen, Wu, Liu, Cheng, "EGR: Equivariant Graph Refinement and Assessment of 3D Protein Complex Structures," *arXiv*, 2022.

PRESENTATIONS

Deep Learning for Science (DL4Sci) School, UC Berkeley / LBNL (Zatom-1)	Jul 2026
LeMaterial Reading Group (Zatom-1, Remote)	Mar 2026
3DSIG at Intelligent Systems for Molecular Biology (ISMB), Lyon & Liverpool	Jul 2023, Jul 2025
Computational Modeling of Molecular Structures (Guest Lecture), University of Missouri	Nov 2024
Structured Probabilistic Inference & Generative Modeling (SPIGM) at ICML, Vienna, AT	Jul 2024
AI to Accelerate Science and Engineering (AI2ASE) at AAAI, Washington D.C.	Feb 2023
Advanced Methods in Deep Learning (Guest Lecture), University of Missouri	Nov 2022
Xuefeng Cui Lab Seminar, Shandong University (Remote)	May 2022

AWARDS & FELLOWSHIPS

Zatom-BIO Genesis Mission Phase 1 Grant, U.S. Department of Energy ($750,000, Proposed)	2026
Hopper Postdoctoral Fellowship, Lawrence Berkeley National Laboratory ($~300,000)	2025 – 2027
Outstanding EECS & CS PhD Student Awards, University of Missouri	2025
Top-10 Reviewer Award (Top 3%), Learning on Graphs (LoG) Conference ($500)	2023
XSEDE Hardware Grant, U.S. National Science Foundation ($500)	2022
Public Outreach Features in *Marktechpost* / *HPCwire*	2021 / 2026

Dean's Engineering Excellence Fellowship, University of Missouri ($10,000) 2020 – 2025
O'Neill Graduate Fellowship, University of Missouri ($10,000) 2020 – 2025
Top of Class Graduate in CS, Mathematics, and Physics, Missouri Western State University 2020

SOFTWARE

Deep Learning

JVP Flash Attention https://github.com/amorehead/jvp_flash_attention
Flash Attention Triton kernel with support for second-order derivatives (20k+ PyPI downloads)

Foundation Models

Nucleus https://github.com/Zatom-AI/nucleus
Central repository for atomistic foundation models with shared components (2.5k+ PyPI downloads)

Zatom https://github.com/Zatom-AI/zatom
General-purpose model for 3D molecules and materials

Molecular Docking & Drug Discovery

FlowDock https://github.com/BioinfoMachineLearning/FlowDock
Geometric flow matching for protein-ligand docking and affinity prediction (CASP16 top-5 method)

PoseBench https://github.com/BioinfoMachineLearning/PoseBench
Comprehensive benchmark for deep learning-based protein-ligand docking (10k+ PyPI downloads)

GCDM https://github.com/BioinfoMachineLearning/bio-diffusion
Geometry-complete diffusion for 3D molecule generation and optimization

AlphaFold 3 - PyTorch https://github.com/lucidrains/alphafold3-pytorch
First open-source implementation of AlphaFold 3 in PyTorch (75k+ PyPI downloads)

Biomolecular Representation & Design

GCPNet https://github.com/BioinfoMachineLearning/GCPNet
Chirality-aware geometry-complete perceptron networks for 3D molecular graphs

MMDiff https://github.com/Profluent-Internships/MMDiff
SE(3) diffusion model for joint sequence-structure generation of nucleic acids and proteins

Geometric Transformer https://github.com/BioinfoMachineLearning/DeepInteract
Geometric deep learning for protein interface prediction (30k+ PyPI downloads)

DIPS-Plus https://github.com/BioinfoMachineLearning/DIPS-Plus
Enhanced database of interacting 3D proteins (40k+ PyPI downloads, integrated into Meta AI's OMol25)

ProteinWorkshop https://github.com/a-r-j/ProteinWorkshop
Benchmarking framework for protein representation learning (15k+ PyPI downloads)

MENTORING

Graduate Students Mentored 2023 – Present

- Miruna Cretu (PhD student at University of Cambridge → Machine Learning Researcher at Boltz)
- Antonia Panescu (PhD student at Yale University)
- Tynan Perez (PhD student at Massachusetts Institute of Technology)
- Ahmed A. Elhag (PhD student at University of Oxford)
- Arun Raja (PhD student at University of Oxford)
- Mahdi Pourmirzaei (PhD student at University of Missouri)
- Zhiyu Wang (MPhil student at University of Cambridge, Highly Commended MPhil Project Prize → Research Scientist at Sapient Intelligence)
- Rylie Weaver (MS student at Claremont Graduate University → PhD student at University of Tennessee, Knoxville)
- Pushpa Kumar Balan (MS student at University of Central Missouri)

Undergraduate Students Mentored 2023 – Present

- Rishabh Anand (BS in CS, National University of Singapore → PhD student at MIT)
- Gautam Bhooma (BS in CS & Chemical Engineering, UC Berkeley)
- Sai Srikar Tummala (BA in Data Science, UC Berkeley)
- Salman Ibrahim Koshy (BS in Operations Research & Data Science, UC Berkeley)
- Sarvagya Goyal (BA in CS & Data Science, UC Berkeley)
- Kyler Zook (BS in CS, University of Missouri)
- Nathan Sekar (BS in CS, Rutgers University)

PEER REVIEW

Organizing Committees: Learning on Graphs (LoG) Conference (2024–2026), ICML ML4LMS Workshop (2024).

Nature Machine Intelligence, Nature Methods, Nature Communications, Transactions on Machine Learning Research (TMLR), Bioinformatics, Briefings in Bioinformatics, BMC Bioinformatics, Journal of Chemical Information and Modeling (JCIM), Computational and Structural Biotechnology Journal (CSBJ), IEEE Transactions on Neural Networks and Learning Systems (TNNLS), IEEE Transactions on Emerging Topics in Computational Intelligence (TETCI), Neural Information Processing Systems (NeurIPS; AI4Sci, AI4D3, GenBio, & MLSB Workshops), International Conference on Learning Representations (ICLR; GEM, AI4NA, & FPI Workshops), International Conference on Machine Learning (ICML; AI4Sci, CB, AccMLBio, & SPIGM Workshops), Association for the Advancement of Artificial Intelligence (AAAI), Learning on Graphs Conference (LoG), ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM-BCB).